PROVISIONAL APPLICATION FOR PATENT

UNDER 35 U.S.C. § 111(b)

INTERPRETATION, SEPARABILITY AND COMBINATION OF CONSENT-GATED SIGNALING, SAFETY GUARD STATE, ARRIVAL CHECK-IN AND SILENT-ARTICULATION SAFETY INPUTS; STANDALONE SOLO, UNENROLLED-PARTNER, NO-MEDIA AND GUARD-STATE MODES; A VOICE RELAY THAT RESTATES A DISTRESSED PERSON'S SPOKEN OR SILENTLY ARTICULATED INSTRUCTIONS; AND A USER-SELECTABLE, ENFORCED SENSING AND INTERPRETATION POLICY

Part D, version 1.0.0 — November 5, 2026.

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Related applications: U.S. Provisional Patent Application Nos. 64/182,340; 64/183,657; 64/189,542; and 64/190,434; U.S. Patent No. 12,347,301 B1; U.S. Provisional Patent Application No. 64/016,800; U.S. Patent Application Publication No. 2016/0196735 A1 (each incorporated by reference)

Entity status: Micro entity (certification on Form PTO/SB/15A filed herewith)

USPTO customer number: 208507

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to U.S. Provisional Patent Application No. 64/182,340, filed October 22, 2026, version 0.9.0, titled "Privacy-Preserving Ambient Consent Signaling Gated on Continuously Renewed Affirmation, with Music, Lapse Detection and Escalation to Help" (the "v0.9.0 application"), which was filed before, and is separate from, Part A; to U.S. Provisional Patent Application No. 64/183,657, filed October 24, 2026 (the "'657 application" or "Part A"); U.S. Provisional Patent Application No. 64/189,542, filed November 2, 2026 (the "'542 application" or "Part B"); and U.S. Provisional Patent Application No. 64/190,434, filed November 3, 2026 ("Part C"), each of the present inventor. The entire disclosure of each is

incorporated herein by reference. Terms defined in those applications have the same meanings here unless this application defines them differently.

[0002] Part A is a separate, later application that restates much of the v0.9.0 application but not all of it. Subject matter of the v0.9.0 application that Part A omits or describes differently remains incorporated here by the reference above. Each rule of section 1 and each statement of section 2 applies to that subject matter as it applies to the corresponding subject matter of Part A. Where the v0.9.0 application and Part A differ, each version is disclosed as an alternative, and Part A does not withdraw any subject matter of the v0.9.0 application.

[0003] This application is written to stand on its own. Where this application states that an element of Part A, B or C may be used alone, omitted, or combined, it restates that element in enough detail that the statement can be understood without reference to the earlier application. Where this application and an earlier application differ, this application governs for its own subject matter and does not narrow the earlier application.

[0004] This application is also related to U.S. Patent No. 12,347,301 B1, titled "Consent management system over a communications network" (the "'301 patent"); to U.S. Provisional Patent Application No. 64/016,800, titled "Networked Emergency Dispatch and Preparedness Simulation System"; and to U.S. Patent Application Publication No. 2016/0196735 A1 of the present inventor (published under the name Adam Clayman). Each is incorporated herein by reference in its entirety.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0005] Not applicable.

TECHNICAL FIELD

[0006] The disclosed subject matter relates to consent management and personal safety systems, and more specifically to systems in which a perceptible output continues only while consent is renewed, to safety guard states armed by ordinary cues, to safety check-ins triggered by context, and to safety inputs given aloud, by touch, by button, or by silent articulation.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a chart of the pairings of non-acoustic speech sensors with the articles that carry or hold them, corresponding to Table 1.

[0008] FIG. 2 is a flowchart of a method of the voice relay.

[0009] FIG. 3 is a flowchart of a method of selecting and enforcing a sensing and interpretation policy.

DETAILED DESCRIPTION

1. Interpretation

[0010] The following rules apply throughout this application and to the subject matter of Parts A, B and C as described here, and are intended to govern any application claiming the benefit of this application.

[0011] *Lists of alternatives.* "At least one of A, B, and C," "at least one of A, B, or C," "one or more of A, B, and C," and "selected from A, B, and C" each mean any one of A, B, and C, or any combination of two or more of them. None requires at least one of each listed item. "A, B, and/or C" has the same meaning. Each listed item is disclosed as an alternative on its own and in each combination with the other listed items.

[0012] *Lists of elements.* Where a system, method, message, record, sequence, or output is described as comprising, including, or having several listed elements, each listed element may be present alone, omitted, or present in any combination with the other listed elements, unless this application states that an element is required for the subject matter to function, including in the minimum operative combinations of section 5.1, which identify for each principal configuration the elements it needs and those that are optional additions. A list introduced by "including," "such as," or "for example" is illustrative and not exhaustive.

[0013] *Selections from several lists.* Where an element is selected from a first list and a related element from a second list (for example, a sensor and the article that carries it, a responder and the method by which the responder is verified, or a source of a musical boundary and a type of boundary), each item of the first list is disclosed in combination with each item of the second list, and with each combination of items of the second list, except where this application states that the two are incompatible.

[0014] *Or.* "Or" is inclusive unless the context requires otherwise.

[0015] *Open terms.* "Comprising," "including," "having," and "containing" are open-ended. "A" or "an" means one or more. "May" and "in some embodiments" introduce options, each of which may be used or omitted.

[0016] *Statutory classes.* A description or claim of a method also discloses a system comprising one or more processors and memory storing instructions that cause the processors to perform the

method, a device configured to perform it, and a non-transitory computer-readable medium storing such instructions. A description or claim of a system, device, or medium also discloses the method it performs. Each step of a method may be performed by any one or more of the devices described, alone or together.

[0017] *Participants.* Single-participant and multi-participant embodiments are each disclosed. Where this application describes a session in which the affirmation of two or more participants is required, it also describes the corresponding single-participant embodiment, in which one enrolled participant's own affirmation renews the affirmation window, as in Solo mode, except where the description requires two or more, as for a joint consent state. Where it describes a single participant, it also describes the corresponding embodiment in which each of two or more participants renews his or her own affirmation window and the joint consent state is the minimum of theirs. A reference to "each participant whose affirmation the session requires" includes a session in which affirmation is required from fewer than all persons present, as in the unenrolled-partner configuration.

[0018] *Outputs.* A reference to media content, music, or playback as the gated output also discloses each other perceptible output described, including a haptic pattern, a lighting state, an ambient tone, the actuation of a device, and the continuation of a recording, except where the context requires audio.

[0019] *Speech inputs.* A reference to a word, phrase, go word, stop word, safe word, safe phrase, or universal safe word detected in speech also discloses that input whispered, sung, hummed where applicable, or articulated silently and detected by a non-acoustic speech sensor as described in Part C, and an equivalent input given by touch, gesture, button pattern, or an alternative modality for a person with a disability, each carrying the same authority, unless the context requires audible speech, except that a silently articulated affirmation is credited only where each participant whose affirmation the session requires has elected to allow silent affirmations, as described in Part C. In Solo mode the enrolled participant is the only participant whose affirmation the session requires, so that participant's own election suffices, and no other person present need agree to silent affirmations or, where applicable law permits and the participant so elects, be aware of them. The gated media then carries the enrolled participant's ongoing consent and its strength: its volume may follow that participant's current consent score, and a fade upon a lapse or a falling score prompts the participant to speak up, take a pause or a brake, give a yellow or red concern cue, or, as needed, allow the escalation sequence to send a distress alert.

[0020] *Exemplary claims.* Claims presented in this application, in the v0.9.0 application, or in Parts A–C are exemplary. A feature recited in a dependent claim may be combined with any independent claim and any other dependent claim with which it is compatible, whether or not the claims as written refer to one another.

2. Separability and combination

[0021] Each feature described below may be used alone, without any other feature of this section, or in combination with any one or more of the others, and with any of the core mechanisms of paragraph [0022]. Where a feature needs another to function, this section says so.

2.1 Core mechanisms

[0022] The core mechanisms are: (a) a gated output that continues only while an affirmation input is renewed within an affirmation window; (b) a gradual change of the output, such as a fade, upon a lapse of the window; (c) an abrupt change of the output upon a stop input, distinguishable from the gradual change; (d) resumption of the output only upon a fresh affirmation; (e) an escalation sequence of check-in, alarm, and call for help upon continued absence of interaction; (f) the distinction between an interaction, which halts escalation, and an affirmation, which alone resumes the output; and (g) a joint consent state computed as the minimum of the participants' consent states.

[0023] Each core mechanism may be used without the others, except as follows. Mechanism (d) presupposes (a). Mechanism (f) presupposes (e). Mechanism (g) is used only where affirmation is awaited from two or more participants. In particular: (a)–(d) may be used without any escalation sequence; (e) may follow a lapse of any gated output, or may be used in a guard state in which no output is gated; and (c) may be used with or without a specified stop latency, a stop latency shorter than the fade period being preferred.

2.2 Elements of the media-control embodiment

[0024] In the embodiment in which the gated output is media played on a media playback target, each of the following may be used or omitted independently of the others and of mechanisms (e)–(g): (i) acoustic echo cancellation of the media from microphone audio using a playback reference signal; (ii) lyric masking or lyric screening, which may be used in place of, or in addition to, echo cancellation, and does not require it; (iii) confirmation, from a player state returned by the playback target, that a fade, stop, or play command took effect, and escalation to a further adapter where it did not; (iv) rendering of the media audibly in the room so that its continuation, fade, and stop are each perceptible without a display; (v) continuous detection

without a wake word; (vi) alignment of a fade to a musical boundary; and (vii) any of the adapters of the media control adapter layer, alone or in any group of two or more.

2.3 Inputs and attribution

[0025] Each of the following input and attribution features may be used alone or with any other, and with any participant configuration of section 2.5: tiered go words, including an enthusiastic tier; the semantic classifier; its asymmetric thresholds; the independent path by which enrolled stop and go words are honored regardless of the classifier; the classification of stop utterances as bare, with an expressed reason, or of limited scope; ducking upon a scoped stop; on-device adaptation of the classifier; the clarifying exchange; reaction scripts; the brake state and brake script; secret concern cues; duress affirmations; the implied-channel analyzer and its tone modes; voiceprint attribution; liveness checks; biometric touch affirmations; device binding, presumed or proven; wearable gestures, including nods, shakes, and hand clasps; button and remote press patterns; the hummed or whistled help motif; universal safe words of a registry; and alternative modalities for a person with a disability.

[0026] In particular, a duress affirmation does not require secret concern cues; the voice relay may attribute the triggering participant's speech by the enrolled device or button that triggered it, without voiceprint attribution; a boundary map may be used by the clarifying exchange, by the robot, and by the breathing-interruption indicator, without the semantic classifier; and the private agent may conduct checklists and disclosures without the semantic classifier.

2.4 Escalation, help, and entry

[0027] Each of the following may be used alone or with any other: repetition of the check-in with increasing intensity; each component of the alarm (a tone heard outside the room, flashing of a lighting device, continuous haptic output, and a spoken warning), any one or more of which may form the alarm; each channel of the check-in (haptic, lighting, spoken, display), any one or more of which may carry it; each election of whether the alarm is audible in the room; each help pathway, in any order; the dispatchable bedroom location; responder entry through an access device; each method of verifying a responder at the door, alone or with another; confirmation of the responder's reported location, which may be required or omitted; the courtesy interval before unlocking, which does not require any particular verification method; the melodic motif signaling that help is on the way, which does not require any particular help pathway; the OK phrase; soft close; and the evidence locker.

2.5 Participant configurations

[0028] Each feature of this application may be used in each of the following configurations, except where the feature requires two or more participants: Duo mode, in which every party present is enrolled; Solo mode, in which one enrolled participant runs the system as a discreet protective layer for himself or herself and the controller processes only that participant's own side of the interaction; an asymmetric session, in which affirmation is awaited only from enrolled participants and the joint consent state is computed over them, while a stop from anyone present is honored; a session with a guest participant enrolled for the session; a session in which no media plays and a silent ambient signal, a lead-time haptic warning, or no continuous signal is used instead; and the guard state, in which no affirmation is awaited and no output is gated, but stop inputs, brakes, concern cues, duress affirmations, button patterns, and distress indicators are honored and the help pathways are armed.

2.6 Policy, relationship, and balance features

[0029] Each of the following may be used alone or with any other: the policy engine and protective derivation of an active policy; per-stage media responses; the discreet mode; stages of intimacy and the gate at the legal line; envelopes; the stage plan; wind-down and the afterglow grace period; cadence-based shortening of the affirmation window; the relationship profile and milestone-based lengthening; new-pair short windows; role preferences, suggestions, and the balance meter; receptive-load weighting; manually recorded initiations and the initiation balance; reward credits and attestations; and gifts. In particular, role preferences, receptive-load weighting, and recorded initiations do not require cadence-based shortening of the affirmation window, and each may be used without the others.

2.7 Bedroom mode and guard state

[0030] Each of the following may be used alone or with any other: learning of a bedroom from charging cycles, wake-sleep cycles, alarm dismissal, wearable-detected sleep, voice detection before and after sleep, or snoring, any one or more of which may be used; assignment of a fixed device to a bedroom in a registration or smart-home application; a place fingerprint recorded by a phone; each bedtime signal and each wake signal, any one of which may enter or leave the bedroom mode; age gating; detection of universal emergency phrases only, enrolled safe phrases only, or both, as the user elects or as a local or institutional policy, a policy of the manufacturer, the copyright author, or the inventor, the terms of a software license, a community safety condition, or applicable law requires; detection from microphone audio, from a non-acoustic speech sensor, from button or touch input, or any combination; offering or entering a guard state upon detecting a second person's device; and each cryptic arming input of the guard state.

2.8 Arrival-triggered check-in (Part B)

[0031] The arrival check-in of Part B is presented when one or more conditions are met. Each of the following conditions may be used alone or with any one or more of the others: (a) the place is unfamiliar; (b) the arrival falls within a configured late-night window; (c) a risk context is present; (d) the place is classified as a residence; and (e) another device not previously associated with the user arrives with the user's device. The conditions may be tested in any order. Any one of conditions (a), (b), (c), and (e) suffices by itself to present the check-in, and condition (d), classification of the place as a residence, is optional, so that the check-in may be presented at a place that is not a residence.

2.9 Silent-articulation inputs (Part C)

[0032] As stated in Part C, each element of the silent-articulation embodiments may be used, omitted, or replaced independently of the others. In addition, a non-acoustic speech sensor may supply any input described in this application, including an arming cue, a go word, a stop word, a brake, a concern cue, a duress affirmation, and an answer to a check-in, subject to each participant's election where Part C requires one.

2.10 Refuge, apology channel, and distancing (Part B)

[0033] *Refuge coordination.* Each of the following may be used alone or with any other: a request made from a device, by a press pattern on a button in a room, or by a greyzone answer to a check-in; screening for medical risk and routing to medical evaluation or a monitored refuge; an inventory of safe-sleep rooms; a registry of homes of vetted hosts; the no-romantic-interest constraint, with or without private set intersection of prior connections, and with or without evaluation in a trusted execution environment; exclusion of refuges in the building, on the floor, or associated with a named person; time-limited credentials for each access device on a path; withholding of the refuge's identity and location; respiration monitoring with consent; arrival verification by handshake or code; a ride for the requester alone; revocation and aftercare at check-out; and the absence of notification to the person the requester left. A safe-sleep room may have any one or more of its described elements: an electronic lock with an inside release, a help button at the center of the headboard, a room tag, a non-imaging presence and respiration sensor, a reporting hub, and the absence of any camera or microphone. Screening for medical risk is optional: a refuge may be selected and credentials issued without it.

[0034] *Mediated apology channel.* Each of the following may be used alone or with any other: the cooling-off period; screening against published criteria, by an operator, by an automated pre-screen, or both; encryption to a key holder the protected party designates, with destruction of the

plaintext; each notification mode; the transmittal record; denial of investigator access; signed read receipts; closure notices or, alternatively, uniform acknowledgments; a comparison of the submission as written and as transmitted; clarifying questions from a closed set; withdrawal notices; limits on the number of submissions; hash verification; a signed hash tree over segments; an append-only log countersigned by an independent witness; counts of submissions by outcome; and operation independent of investigation and adjudication. Each may be used for a no-contact order that is one-way or mutual. The cooling-off period is optional: a submission may be screened and delivered without it.

[0035] *Privacy-preserving distancing.* Proximity tokens derived from a pair secret, the oblivious lookup of protected zones, and the suggestion to change direction, route, or timing may each be used alone or with the others, and with the refuge and apology features.

2.11 Voice relay (the megaphone)

[0036] *Trigger and attribution.* The voice relay of Part A [0175] repeats, through a loudspeaker in the room, what is said by the person who triggered it, so that the person's request is heard clearly and with emphasis. The relay may be opened by any one or more of a stop input, a brake input, a concern cue, a duress affirmation (only where the person specifically elected in advance that a duress input open an audible relay, as stated in section 5.4), a button pattern, an enrolled megaphone word, a request in the person's own words, and a silently articulated input. It relays only the triggering person. That person is identified by voiceprint attribution, by the enrolled device, wearable, or button that raised the trigger, or by a non-acoustic speech sensor worn by or directed at that person, any one of which suffices. Speech of any other person is never relayed.

[0037] *Silent input voiced aloud.* Where the triggering person articulates an instruction silently and it is detected by a non-acoustic speech sensor as described in Part C, the relay voices the instruction aloud. A person who cannot or does not speak above a whisper, for example during a freeze response, can thereby make a refusal or request audible to everyone present.

[0038] *How the restatement differs from the utterance.* The restatement may differ from what the person said in any one or more of amplitude, voice, and phrasing. Each of the following may be used alone or with any other:

- *Amplitude.* The restatement is rendered at an elevated volume, and the volume may rise while the relay stays open, upon repetition of the same instruction, or where the instruction is not followed, as sensed by continued interaction, the absence of separation, or a further stop or concern input.

- *Voice.* The restatement may be rendered in a synthesized voice whose characteristics the person selects in advance, including pitch, timbre, apparent gender, apparent age, accent, cadence, and firmness. It may instead use a voice the person chose or recorded in advance, or, where the person elects, a replay of the person's own captured audio. By default the synthesized voice is used, so that the person's own voice is not broadcast.
- *Phrasing.* The restatement may be verbatim; a paraphrase in plain, direct language; a summary condensed to the operative request; or a restatement as a clear, firm, specific instruction phrased to make compliance more likely. Such a restatement names the action requested, such as stopping, stepping back, letting go, getting dressed, or leaving the room. It may address the other person directly, remove ambiguity, and state that help has been or will be called where that is so. The phrasing is chosen on the device from the recognized request, within forms the person approved in advance, and never adds a request the person did not make.

[0039] *Other elements.* Each of the following may be used alone or with any other: a short distinctive tone before each restatement; a private confirming question where recognition confidence is low, with only confirmed requests relayed; advancement to the help stage upon a confirmed request for help; keeping the relay open, relaying only the triggering person, until the persons separate or that person closes it; and rendering through any loudspeaker present, including a phone, a smart speaker, an attending device, or a venue's room system. The relay may operate in a consent session in Duo or Solo mode, in a guard state, in a bedroom mode, and outside the bedroom. Whether relayed speech is also transcribed or retained is governed by the recording settings and not by the relay. Activation of the relay is support for the person who invoked it and is not a finding of fault.

[0040] *FIG. 2.* FIG. 2 illustrates a method 200 of the voice relay. At step 202 a trigger of [0036] is received from a person. At step 204 the triggering person is identified by voiceprint attribution, by the enrolled device, wearable, or button that raised the trigger, or by a non-acoustic speech sensor worn by or directed at that person. At step 206 a device recognizes an instruction spoken or silently articulated by that person, relaying no other voice. Where recognition confidence is below a threshold at step 208, the device asks the person a confirming question privately at step 210 and relays only a confirmed instruction; where the person declines or does not answer, the device relays nothing at step 220, continues to listen, and, where the policy so provides, advances the escalation sequence. At step 212 the device selects the amplitude, voice, and phrasing of the restatement within the forms the person approved in advance. At step 214 the device renders the restatement through a loudspeaker, preceded by a distinctive tone. The relay remains open at step 216, relaying further instructions of the same

person and raising the volume, within the bounds of section 5.4, where an instruction is not followed, until at step 218 the persons separate or the person closes the relay. The relay may omit steps 204 and 208 to 212 only where the embodiment otherwise satisfies the conditions they serve, for example by isolating the person by a personal sensor or by rendering only a fixed announcement. The steps may be performed by one device or divided among several.

3. Risk-context signals restated

[0041] Part B's risk-context engine refers to several mechanisms of Part A. They are restated here so that the arrival check-in and its risk context can be understood from this application alone. Each signal is optional, each is processed on the device, and any one or more may be used. A user may instead elect a simple rule that removes the risk-context test entirely, such as checking in upon every late-night arrival at an unfamiliar place.

[0042] *Leaving-the-party breath reading.* A user may enroll a rule that, upon leaving a venue or event at which the user has indicated alcohol was consumed, the device asks for a breath sample on an enrolled breath-alcohol sensor. Departure is detected by the user's own check-out, by a venue's exit station, by geofence departure, or by a scheduled prompt. The reading is time-stamped, held on the device and in the user's own encrypted record, and never transmitted to a partner or platform. A reading at or above a configured level, for example 0.04 g/dL within the preceding six hours, is a risk-context signal. The same reading may separately apply a consent lockout as described in Part A.

[0043] *Inferred drinking episode.* An on-device model may infer a drinking episode from the phone's and wearable's sensors, such as movement, gait, and the time and place pattern of the evening.

[0044] *Departure from a drinking occasion.* Departure from a bar, club, registered party, or other drinking occasion, detected as in [0042], is a risk-context signal whether or not a breath sample is given.

[0045] *Indication of a date.* A date is indicated by a calendar entry the user marked as a date, or by the dating-platform integration of Part A. In that integration, upon mutual opt-in by two matched users, the platform passes a pseudonymous match token to each user's device. When the two meet, their devices perform a meeting handshake over a local link, by Bluetooth Low Energy or ultra-wideband ranging confirming that two devices holding the same match token are within a short distance, or by one device scanning a code displayed by the other. The handshake binds the match token to each enrolled user and to the device present. A completed handshake

earlier in the evening is an indication of a date. The platform receives at most an outcome signal and never receives audio, enrolled words, consent states, locations, or voiceprint data.

[0046] *Co-travel.* Another device having remained near the user's device from the place of departure to the place of arrival is a risk-context signal. Co-travel may be indicated by a meeting handshake as in [0045], by an exchange of ephemeral tokens between two devices running the system, or by the persistence over the trip of short-range wireless signals of a single nearby device, judged within short windows without identifying that device or retaining its identifiers. Devices of the user and of the user's chosen contacts are excluded.

[0047] *Other signals.* A ride that ended at the arrival place rather than at the user's home, or a shared ride on which another rider's stop was skipped; a night-out mode the user turned on earlier; a request from a chosen contact that the user's devices offer a check-in, which may be sent as a touch message composed on a wearable as a tap pattern, a drawn sketch, or a transmitted heartbeat; and an impairment flag raised from the user's own voice, from slurred or fragmented speech or inconsistent answers, or from a breath reading.

[0048] *Combination.* The engine may combine the signals into a score, for example a weighted sum compared with a threshold, or may treat any one signal as sufficient. No signal is sent off the device for scoring.

4. Pairings

4.1 Non-acoustic speech sensors and the articles that carry them

[0049] Table 1, which FIG. 1 also shows, discloses each pairing of a non-acoustic speech sensor of Part C with an article that carries or holds it. "Stated" marks a pairing Part C [0030] already describes. "Disclosed" marks a pairing first disclosed here. A dash marks a pairing not illustrated in this table; a dash is not an exclusion, and other compatible pairings may be used. Each marked pairing may be used alone, with any other marked pairing, and with a microphone. Each pairing feeds a sensor-specific front end as described in section 5.2.

Article	RF antennas	Radar	Surface EMG	Ultrasonic	Inertial or vibration	Ear-canal	Optical non-imaging	Oral contact
Eyewear (temples, hinges, nose pads, bridge)	Stated	—	Disclosed	Disclosed	Disclosed	—	Disclosed	—
Earbud or hearing aid, including the stem	Stated	—	Disclosed	Disclosed	Stated	Stated	Disclosed	—
Band behind the ear	—	—	Disclosed	—	Disclosed	—	—	—
Necklace, collar, choker, or neckband	Stated	—	Stated	—	Stated (laryngeal)	—	Disclosed	—
Headband, cap, or sleep mask	Stated	—	Stated	—	—	—	—	—
Mouthguard, night guard, or retainer	—	—	—	—	Disclosed	—	—	Stated
Bedside sensor (not worn)	Stated	Stated	—	—	—	—	—	—

[0050] Each sensor of Table 1 may supply each input of [0032] in each placement marked for it. Two or more sensors in one article, or in two or more articles, may be used together, agreement between them raising confidence. For a stop or help input, a detection in any one above its threshold suffices to prompt protective action; an affirmation or a cancellation requires the attribution and authorization it would require from a single sensor. Each placement may hand the guard state to any other placement, to a press, tap, or squeeze pattern on a ring, watch, or wearable button, or, where enabled, to a microphone, when the article carrying the sensor in use is taken off.

4.2 Responders, verification, and entry

[0051] Each authorized responder type described in Part A (a security professional, a monitoring service, a neighbor, a roommate, a friend, a family member, a building doorman, concierge, or receptionist, a peer responder, an emergency medical technician, a paramedic, a firefighter, a police officer, and an emergency service) is disclosed in combination with each of the following, alone and in any combination: (a) designation as permitted or not permitted to be admitted; (b) each method of verifying the responder at the door, namely a one-time code, a credential presented by the responder's device, the building's access-control system, a monitoring operator, and an audibly verified knock; (c) confirmation, or no confirmation, of the responder's reported location; and (d) each access device, namely a smart lock, a building intercom or access-control system, a gate or garage controller, a venue door system, and, for a door with no access device, entry instructions or a keyholder neighbor.

4.3 Musical boundaries

[0052] Each source of boundary times described in Part A (section or audio-analysis metadata of a streaming service, time-synchronized lyrics, a queue and track-duration state of a media player, and on-device beat tracking or structural segmentation of a playback reference signal) is disclosed in combination with each type of boundary (the end of a bar, a phrase, a chorus, a verse, a bridge, or a lyric line within a track, and a boundary between tracks), and with each action aligned to a boundary (the start of a fade, a crossfade to a prompt track, the start of a next track, a resumption, and a renewal reminder).

4.4 Other lists

[0053] Each transport for a call for help described in Part A may be used alone or as a fallback after any other. Each alternative modality for a person with a disability may replace each output or each input it is listed for. Each venue-owned device may conduct a session alone or with any other. Each arming cue may arm the guard state, the bedroom mode, or a consent session, alone or with any other arming cue.

4.5 Combinations of the claims of Parts B and C

[0054] Each of claims 31 to 40 and 48 to 56 of Part B is disclosed in combination with each other one of them and with claim 30 of Part B. Each of claims 16 to 26 and 47 of Part B is disclosed in combination with each other one of them and with claim 15 of Part B. Each of claims 2 to 11 and 44 to 46 of Part B is disclosed in combination with each other one of them and with claim 1 or claim 12 of Part B, and with claim 7 of this application. Claim 58 of Part B is disclosed with claim 57 of Part B, and claims 15 to 26 and 43 of Part B are each disclosed with claim 27 of Part B and with claim 7 of this application. Each of claims 2 to 15 of Part C is disclosed in combination with each other one of them, with claim 1 of Part C, and with claims 5 and 6 of this application. Each of claims 17 and 18 of Part C is disclosed with the other and with claim 16 of Part C, and each of claims 26 and 27 of Part C with the other and with claim 25 of Part C. In each case the combination excludes only features stated to be alternatives.

5. Principal combinations: minimum configurations and implementations

5.1 Minimum operative combinations

[0055] This section identifies, for each principal configuration, the elements it needs in order to function and the elements that are optional additions. The rule of [0012] that listed elements may be omitted is subject to these minimum combinations. Each configuration is described affirmatively below and may be combined with any other.

[0056] *(a) Consent-gated output.* An input device produces affirmation events, each attributed to a participant, and stop events, which are honored whether or not they are attributed. For each participant whose affirmation the session requires, an affirmation timer is reset only by that participant's affirmation events. An output controller continues the output only while every such timer is current and no stop event has been received, changes the output gradually upon a lapse, and changes it abruptly upon a stop. Optional additions include the escalation sequence, the semantic classifier, voiceprints, musical boundaries, and the policy engine. Where an escalation sequence is used, its controller preferably runs independently of the output controller, so that a failure to control the output does not prevent the check-in, alarm, or call for help.

[0057] *(b) Guard state.* An arming input enters a guard state in which no affirmation is awaited. A safe-input detector runs in the guard state. Upon a safe input, a check-in presenter presents a private check-in, and a help controller initiates a help response upon a help answer, a duress answer, or no answer. Optional additions include covert arming, silent sensors, and a tiered help response; alternatives include immediate help without a check-in (section 5.7).

[0058] *(c) Silent safety input.* A non-acoustic speech sensor worn by or directed at the user feeds a sensor-specific front end. A recognizer produces a safety event carrying an event type, an attribution to a person, and a confidence. A safety controller (the guard state, a consent session, or the voice relay) acts on the event. Optional additions include fusion with a microphone and a second sensor.

[0059] *(d) Covert duress.* A duress-input recognizer records duress in an internal safety state and invokes a help pathway, while a presentation controller holds every output perceptible to another person in the state an ordinary input would have produced (section 5.3).

[0060] *(e) Voice relay.* A trigger authorizes a relay owner. The owner's subsequent instructions are isolated from other sound (section 5.4). A restatement generator produces a restatement whose meaning is checked against the instruction (section 5.5). The restatement is rendered through a loudspeaker or to a communications endpoint, with the relay's own output rejected from its input.

[0061] *(f) Arrival check-in.* An arrival detector feeds a condition evaluator, any one condition of which suffices. A private check-in is presented, and the answer is acted upon, which may be a local action only.

[0062] *(g) Monitoring handoff.* Two or more input channels report to a handoff controller that holds the pending check-in, timers, and escalation state independently of any channel (section 5.6).

[0063] *(h) Sensing and interpretation policy.* A selector maps each position to a machine-readable processing policy. A policy enforcer applies the policy to each device and confirms that it took effect (section 5.8).

5.2 Recognizing silent safety inputs

[0064] Silent safety inputs may be recognized in three ways, each usable alone or in combination. They differ in what is recognized, and the system may run more than one at once.

[0065] *Enrolled-cue recognition.* The user enrolls a small set of cues, such as a silent safe phrase, a pattern of tongue clicks, or a jaw clench, by performing each several times during a rehearsal, for example three to ten times. The front end filters the sensor signal, for example by band-pass filtering and removal of drift and motion artifacts. It segments candidate events by the onset of activity and extracts features. Each candidate is compared with the user's enrolled templates, for example by dynamic time warping or by the distance between embeddings computed by a small on-device model. A detection requires a similarity above a per-cue threshold. The threshold is set during calibration from recordings of the user's ordinary movements, such as chewing, swallowing, yawning, speaking, and kissing, so that those movements are rejected. A cue that calibration shows to be confusable with another cue or with ordinary movement is flagged, and the user is asked to choose another.

[0066] *Fixed safety vocabulary.* A recognizer trained in advance recognizes a closed set of safety intents, for example stop, wait, slow down, not that, I'm okay, help, step back, let go, and call a named contact. An intent may carry arguments drawn from a closed set, such as a body region, an act, or the name of a designated contact. The recognizer may be personalized from a few renditions by the user. An utterance outside the vocabulary yields an unknown result, on which no action is taken other than, where the policy so provides, a private check-in.

[0067] *General silent-speech decoding.* In a preferred embodiment, general decoding runs only after a conflict signal (section 5.9) or where the participant has expressly elected it; the enrolled-cue and fixed-vocabulary embodiments generate no transcript. A sequence model maps sensor features to text or to phonetic units, and the text is then interpreted into an intent. The model may be adapted to the user. Its output carries a confidence for each word and for the intent. Below a confidence threshold, the system falls back to the fixed vocabulary or to a private confirmation. A decoding below the threshold never cancels or downgrades a pending check-in or help response.

[0068] *Sensor-specific front ends.* Each sensor family measures a different physical signal:

- *RF antennas* (Part C): the amplitude and phase of transmission or reflection between antennas near the face, which vary with the positions of the tongue, lips, and soft palate, sampled across one or more frequencies.
- *Surface EMG*: electrical activity of facial, jaw, or neck muscles, sampled at, for example, about 1 kHz per channel, from which envelope and spectral features are computed.
- *High-frequency acoustic or ultrasonic sensing*: probe signals emitted by a speaker and received by microphones of eyewear or an earbud, whose echo profiles change with lip and jaw movement. The signals may be at the upper edge of human hearing, for example about 18–20 kHz, or, in an ultrasonic embodiment, above about 20 kHz, chosen within the limits of the transducers and at levels comfortable for the persons present and for nearby animals.
- *Inertial or vibration*: accelerometer and gyroscope signals from the jaw, cheek, ear, or throat, reflecting jaw rotation and skin vibration.
- *Ear-canal*: deformation of the ear canal with jaw movement, measured acoustically or by a pressure sensor of an earbud.
- *Optical non-imaging*: reflected infrared intensity or photoplethysmographic signals from skin over the lips, cheek, or throat.
- *Oral contact*: capacitive or pressure contacts between the tongue or teeth and an oral appliance.
- *Bedside radar or RF*: range, phase, and micro-Doppler signatures from the region where the user's head lies.

In one embodiment, each front end outputs a fixed-length embedding at a regular rate, for example every 20 milliseconds, and the embeddings of all available sensors feed a shared intent model. In another embodiment, each sensor has its own recognizer, and their outputs are fused at the level of intents. The figures given are examples; no particular recognition accuracy is asserted.

[0069] *Wearer-selective sensing and interference.* A worn sensor responds chiefly to its wearer's articulators because of its contact with or proximity to the wearer. It can nonetheless pick up external disturbances, such as a partner's touch, movement of the bed, or vibration conducted from another person. A bedside sensor may sense more than one person. The system therefore attributes a detection to the wearer only where it is consistent with the wearer's enrolled patterns and, where available, is corroborated by on-body detection or a second sensor. It treats large external motion sensed by an inertial sensor as an indicator of interference: a detection coinciding with such motion is acted upon when corroborated, and otherwise results in a private

check-in, so that a genuine safety input given during a struggle is not discarded. For a bedside sensor, it attributes by the position of the source and the enrolled pattern. It treats an unattributed or ambiguous detection by presenting a private check-in rather than acting as if the detection were attributed. The description in Part C of a sensor that senses substantially no one else's articulators is to be read with these qualifications.

5.3 Internal safety state and presented state

[0070] The controller maintains two states independently. The *internal safety state* records what the system knows and is doing: the consent state of each participant, lapses, stops, brakes, scoped requests, duress, distress, pending check-ins, the escalation stage, and help requests. The *presented state* governs every output perceptible to a person other than the user, including media, lighting, sounds, and announcements. The presented state ordinarily follows the internal safety state. The two may diverge only in the covert cases of Table 2, and only to conceal a protective action. The internal safety state never credits a duress input as an affirmation and never cancels a pending help request because of what is presented, and each divergence is recorded in the user's sealed log.

[0071] Table 2 states the precedence among events and their effect on each state.

Event	Internal safety state	Presented state	Precedence
Lapse	Lapse recorded; escalation timing begins	Gradual change; no change in a discreet mode	Yields to stop, duress, and explicit help
Enrolled stop	Withdrawn; any pending affirmation void	Abrupt change within the stop latency	Overrides lapse, scoped request, brake, and continuation; not overridden by a classifier
Scoped request	Greyzone; does not renew affirmation	Ducking, announced in the speaker's words	An enrolled stop word in the same utterance controls
Brake	Brake state, a cooperative pause	Fade and brake lighting	Yields to stop, distress, and explicit help
Duress, in a session	No affirmation credited; duress help pathway invoked	Unchanged, as for an ordinary affirmation	Overt announcement or relay only on the user's specific prior election
Duress, in a guard state	Distress event; help pathway invoked; no timer involved	Unchanged	Same as above
Explicit help	Help stage	According to the alarm election	Overrides check-in and lapse handling
No input channel	Degraded protection; escalation stays armed	Private notice to the user only, unless elected	Does not end the session or guard state
Interaction before a check-in	Halts advancement of an unattended-session timer	Output not resumed	Interaction attributed to a participant qualifies; incidental movement does not, unless the policy so designates
Response to a directed check-in	Halts advancement only for that check-in	Output not resumed	Only a qualifying response of the participant to whom the check-in is directed
Any interaction after duress or explicit help	No effect on escalation	Unchanged	Ordinary conversation or another person's activity never halts duress or help escalation
Stop and duress together	Both recorded; duress help pathway invoked	The stop is shown; the duress is not	The stop controls what is presented; the duress controls the escalation

[0072] In a discreet mode, a lapse produces no overt change, but a stop still changes the gated output abruptly. Where the gated output is itself covert, such as a haptic pattern perceptible only to the participants, the abrupt change is made in that output. In a guard state, a duress input is treated as a distress event that invokes a help pathway; it does not renew or imply any affirmation timer.

5.4 Voice relay: authorizing the owner and isolating the owner's speech

[0073] Identifying who triggered the relay does not by itself identify which later sounds came from that person. The relay therefore isolates the owner's subsequent instructions by one or more of the following, each usable alone or with the others:

- a personal near-mouth or contact sensor of the owner, such as a non-acoustic speech sensor, a bone-conduction microphone, or an in-ear microphone, which supplies the instruction directly;
- an enrolled speaker model selected upon the trigger and used to verify each later utterance;
- a microphone array that localizes the owner at the time of the trigger, from the source of the trigger utterance or the position of the owner's wearable determined by ranging, and tracks that source by beamforming, rejecting sound from other sources below a confidence threshold;
- an instruction window, opened by the owner, for example by holding a button, and indicated privately by a haptic pulse, during which a short utterance is accepted if it also passes speaker verification or comes from the owner's personal sensor; and
- where speech overlaps or attribution is ambiguous, relaying nothing of the ambiguous segment and instead asking the owner privately or rendering only a fixed announcement.

[0074] *Fixed announcement without biometrics.* A person who has enrolled no speaker model or personal sensor may trigger, by a button or a universal safe phrase, a fixed announcement, for example "A stop has been requested. Please stop and step back." The announcement restates nothing the person said and needs no attribution beyond the trigger.

[0075] *Rejecting the relay's own output.* The relay's synthesized output and any media being played are supplied as references to echo cancellation and to the speaker verifier, so that the relay does not relay its own output or another device's playback. A recognized input is discarded as playback only on evidence of playback origin, such as correlation with the reference signal, timing consistent with the output path, or attribution to the loudspeaker's position. A live repetition by the owner, for example the owner saying "stop" again right after the relay does, is relayed or acted upon when independently attributed to the owner.

[0076] *Two relays.* Where two persons trigger relays at overlapping times, each relay isolates its own owner. Restatements are queued so that they do not overlap, each is preceded by its tone, and a stop takes precedence over a request of lower urgency. Where both are stops, both are rendered.

[0077] *Ending and transfer.* Authorization ends when the owner closes the relay, when the persons are sensed by ranging to have separated, when a configured maximum duration, for example ten minutes, passes without a further instruction, or when the owner's device or sensor is no longer bound to the owner. Authorization does not pass to another person except by that person's own trigger.

[0078] *Bounds on volume.* The relay begins at a level above ordinary speech and may increase in steps, for example of 3 dB, up to a configured maximum. The maximum is set below a level unsafe for hearing at the expected distance and below any maximum the owner sets. Increases are made only upon repetition of the instruction by the owner or upon evidence that the instruction has not been followed. Such evidence is continued proximity sensed by ranging between the persons' devices; contact corroborated by a two-position bed sensor, radar, or the owner's wearable; a further stop, brake, or concern input; or the owner's private confirmation. Movement alone is not treated as such evidence.

[0079] *Duress and the relay.* A duress input opens an audible relay only where the owner specifically elected in advance that it do so, because an audible relay would reveal what the duress input is designed to conceal.

5.5 Restating an instruction without changing its meaning

[0080] The restatement generator preserves the owner's meaning by the following procedure. First, it decodes the recognized instruction into structured fields: an action (such as stop, wait, release, step back, leave, or call), an addressee, an object or body region, a scope (one act or everything), a polarity (do or do not), and an urgency. Second, it selects an approved template for those fields or generates a candidate restatement on the device. Third, it decodes the candidate again and checks that the action, scope, and polarity are unchanged and that the candidate contains no demand or factual assertion absent from the instruction. Fourth, where the check fails, it renders an approved literal form, asks the owner privately, or declines to paraphrase and renders the instruction verbatim.

[0081] *Examples.*

- "Stop touching my arm" may become "Stop touching her arm now," but not "Stop everything," because the scope is the arm.
- "Don't call the police" keeps its negative polarity. It is never rendered as a statement that the police will be called.
- "I need space" may become "She needs space. Please step back," but not an instruction to leave the building.
- "Help is on the way" is rendered only after a responder or service has accepted the request or confirmed a response (section 5.6). Before then, the relay may say only that a request has been sent or delivered.

[0082] The restatement may refer to the owner in the third person or address the other person directly, as the owner elected. The language of the restatement may differ from that of the instruction; the owner may elect, for example, the other person's language, and the meaning check operates on the structured fields, independently of language. In a fixed-vocabulary implementation, each intent maps to restatements approved in advance in each supported language, so that no generation is needed.

5.6 Monitoring handoff and delivery of help

[0083] *Planned handoff.* When an input channel is about to cease, for example because a worn sensor is being removed or a battery is low, the handoff controller identifies a replacement channel and requests its readiness. The replacement acknowledges readiness only after its sensor is active and its recognizer is loaded. Both channels then run concurrently for a short overlap, for example a few seconds. Because independently detecting sensors will not assign the same identifier to one real-world input, the controller correlates their events by participant, event type, and overlapping time window and assigns a canonical event identifier, suppressing duplicates. Only then is the old channel released.

[0084] *Abrupt loss.* When an input channel ceases without warning, for example through hardware failure or a lost link, no overlap is possible. The controller declares degraded protection at once, activates a replacement channel where one is available, preserves the pending state, and records the interval without coverage in the event log. Pending check-ins, timers, and the escalation stage are held by the controller rather than by any channel, so they survive a handoff and are restored from the event log after a restart. Where no channel remains, the controller enters a degraded-protection state, notifies the user privately, keeps escalation armed and, where the user so elected, treats the loss as a lapse.

[0085] *Delivery states.* Each emergency event carries an event identifier and moves through the states queued, transmitted, delivered, accepted, and resolved. Retries across transports reuse the identifier, so that recipients and services suppress duplicates. Where delivery is not confirmed within a timeout, the controller advances to a further transport. An indication that help is on the way, including the acknowledgment motif, is rendered only upon the accepted state. A cancellation by the user is sent as a message referencing the identifier. It never withdraws a request that a responder has accepted; that responder is told of the cancellation and may still check on the user as the policy provides.

5.7 Components and alternative arrangements

[0086] Each mechanism may be supplied by a separate component, and the operations of each are described separately. A *wearable* may sense, recognize, sign, and transmit attributed safety events without implementing any response. A *phone or SDK* may perform attribution, apply policy, run timers, and present check-ins. An *output controller*, such as a playback device, smart speaker, lock, or light, may consume authenticated events, verifying their signature, counter, and freshness as in the renewal-token format of Part A, without implementing any sensor. A *service* may relay help requests and track their delivery, acceptance, and cancellation.

[0087] *Alternatives.* The guard state may be armed explicitly, covertly, or automatically by context, such as a bedroom mode, an arrival check-in, or detection of a second person's device. A safe input of a help category may initiate help immediately, without an intervening check-in. A response may be local only, such as stopping media, a pretext call or message, or a relay announcement, without contacting anyone. The voice relay may operate outside any room and independently of any consent session, rendering through a phone speaker, a remote smart speaker, or a call to a communications endpoint such as a designated contact's phone.

5.8 User-selectable sensing and interpretation policy

[0088] Part A describes a listening ladder, selected by each participant, of rungs L0 (no listening) through L5 (on-device semantic analysis during an active session), with immediate reductions, authorization before increases, session-scoped higher rungs, and a return to a resting rung. This section describes a selector and the mechanisms by which the chosen level is enforced. They may be used with the ladder or with any other set of levels.

[0089] *Selector to enforced policy.* A slider, segmented control, menu, physical switch, or voice command presents positions, each mapped to a machine-readable processing policy. The selector need not be organized as a scale. It may instead be a set of checkboxes or toggles, a grid of features by participant or by device, a list of presets with individually adjustable features, or a conversational or voice dialog, by which the participant selects any combination of individually selectable privacy features; the selected combination is then mapped to a machine-readable processing policy in the same way as a position. Combinations that the policy enforcer cannot satisfy together, or that are disallowed by a required policy, are shown as unavailable with the reason. The policy specifies which sensors and recognizers operate; which enrolled and universal words, phrases, and articulatory cues are recognized; whether transcription, semantic interpretation, and affect analysis are permitted; whose inputs may be processed; and when an authorization expires. Recording, transmission, biometric attribution, and affect analysis are each authorized separately. A high position does not by itself authorize any of them. In a preferred

embodiment, no position permits transcription, diarization, or conflict recording before a conflict signal; those functions are available only as section 5.9 describes.

[0090] *Presets and independent controls.* One control may choose a preset bundle, such as button only; button and safe phrase; safe phrase and universal stop words; adding breath-alcohol readings and the arrival check-in; adding the guard state; and adding the consent envelope. Alternatively, the user may separately choose sensing, interpretation, retention, and sharing. The presets need not form a strict scale of speech analysis. A preset that adds breath-alcohol testing, for example, does not thereby authorize more microphone processing, and the consent envelope may run entirely on button inputs.

[0091] *Changes as sets of permissions.* Because a change of position may remove some permissions and add others at once, for example disabling transcription while enabling location sharing, the system compares the old and new policies permission by permission. Removed permissions are revoked at once, and transient data no longer permitted is discarded. Added permissions are first previewed, showing what will be newly processed, which participants are affected, which capabilities become available or unavailable, and whether anything will be retained or transmitted; the preview activates nothing, and each added permission takes effect only upon authorization by each participant whose inputs it would newly process. Unchanged permissions remain in force.

[0092] *Participants at different levels.* Where participants choose different levels, the processing of each participant's inputs is held to that participant's own level by one or more of: separate personal input channels; attribution of each segment to a participant before any more expansive processing stage; or, where reliable separation is unavailable, limiting a shared microphone to the most restrictive level chosen by any participant whose speech it may capture. The system never transcribes everyone first and discards unauthorized transcripts afterward. Where a participant's level includes universal stop phrases, their recognition is not a permission to analyze anyone's conversation. Whether a level includes universal stop phrases is an option of that level. As alternative policies, a level may instead limit recognition to enrolled phrases only, or to wake-word operation as in Part A's L1 rung. Recognition of universal stop phrases is nonetheless required, and cannot be turned off by the participant, where a local or institutional policy, a policy of the manufacturer, the copyright author, or the inventor, the terms of a software license, a community safety condition, or applicable law so requires. The interface states which phrases each level recognizes and when universal stop recognition is required.

[0093] *Silent-articulation levels.* Because a non-acoustic speech sensor may remain active when the microphone is off, the selector offers separate choices for articulation: no articulation

recognition; enrolled articulatory cues only; enrolled silent phrases; a limited safety vocabulary; and general silent-speech decoding with instruction interpretation. The interface states which channels remain active at each position.

[0094] *Observable enforcement.* The policy enforcer disables processing modules and subscriptions that the policy does not permit and propagates the policy, under a version identifier, to each participating device. It awaits each device's acknowledgment within a bounded timeout. The interface distinguishes the requested configuration from each device's confirmed configuration, and shows a device that has not acknowledged as unconfirmed rather than assuming it complied. Each device holds its processing authorizations locally with an expiry, so that a device that cannot be reached stops the expanded processing when its authorization lapses. Where a device cannot comply, the enforcer places it in a configuration that remains within the processing authorizations then in force, even if a safety function consequently becomes unavailable, and shows the user the reduced protection. Session-scoped authorizations revert to the resting level when the session ends. Each preset may be associated with an allowed-egress manifest, and actual transmissions are compared with it.

[0095] *Separate authorities.* Authorization to process inputs is separate from authority to cancel a pending help response, which remains governed by the user's cancel settings. Activation by the user, escalation after an unanswered check-in, recording, and transmission are each permitted separately, so that a level permitting escalation upon non-response does not by itself permit recording.

[0096] *FIG. 3.* FIG. 3 illustrates a method 300 of the selector. At step 302 a participant changes the selection, by moving a slider to a position or by checking or unchecking individual features. At step 304 the system compares the old and new policies and determines which permissions are removed, added, and unchanged. Removed permissions are revoked at once at step 306. Added permissions are previewed at step 308, and at step 310 the system determines whether each affected participant has authorized them; absent authorization, the added permissions stay inactive at step 312. At step 314 the versioned policy is propagated to each device, and at step 316 acknowledgments are awaited within a bounded timeout. At step 318 each device that did not acknowledge is marked unconfirmed, relies on its expiring local authorizations, and is placed in a fallback within the authorizations in force. At step 320 the confirmed configuration of each device is displayed beside the requested configuration.

5.9 Transcription only upon a conflict signal

[0097] In a preferred embodiment, the system does not transcribe speech, diarize it, or record it unless there is a clear signal of a conflict. Until such a signal, every input is recognized by on-

device keyword spotting, template matching, or classification into a closed set of events, none of which generates a transcript. No audio or sensor signal is retained beyond a short rolling buffer held only in volatile memory, for example of at most 30 seconds, over the channels the participant's policy authorizes. The buffer is overwritten continuously and discarded unless a conflict signal arrives, and its contents may be sealed into a conflict recording only where the policy and applicable law permit pre-trigger content to be kept. This preferred embodiment may be used with every configuration of this application; the elected rungs of Part A's listening ladder that permit on-device transcription during a session remain available as alternatives.

[0098] *Conflict signals.* A conflict signal is an operational trigger category, not a determination that any conflict or misconduct occurred. It is any one or more of a stop input, a brake input, a concern cue, a distress indicator, a duress input, a request for help, a greyzone answer to a check-in, and a trigger of the voice relay, given by any participant by any modality, including silent articulation. Upon a conflict signal, and only as the active policy, each affected participant's selected policy, and applicable law permit, one or more of the following begins: a conflict recording, which may include the contents of the rolling buffer; diarization, to attribute speech to the persons present; and transcription. Which of these begins is set in advance by the participant and governed by the conflict-recording settings and jurisdiction table described in Part A.

[0099] *Bounds.* Transcription and diarization are limited to the participants whose selected policy permits them, under section 5.8. They end when the conflict is resolved, for example by an all-clear answer or the close of the session, or when a configured period expires, whichever comes first. Where an authorized participant and a participant who has not authorized recording share an audio channel that cannot be separated, the channel is recorded only within the narrower of their authorizations or as applicable law expressly permits. Output is encrypted so that each participant can decrypt only what he or she is authorized to receive, for example by separate per-participant copies or per-person keys over shared segments. Recognition by the voice relay after its trigger need not produce text: in the fixed-vocabulary and direct signal-to-intent embodiments, the relay recognizes an intent without generating any transcript.

EXEMPLARY AND ILLUSTRATIVE CLAIMS

The following claims are exemplary and illustrative and do not limit this disclosure or any claim later presented in an application claiming its benefit. Claims 1 to 8, 79, 91, 95 to 97, 99 to 102, and 106 are independent method or device claims, and claims 114 and 116 are system and medium claims. Each dependent claim recites one feature and refers to every independent claim with which that feature is compatible, so that each feature is disclosed with each such independent claim and, through [0020], with each other compatible dependent claim.

Independent claims

1. A method performed by one or more computing devices, comprising: rendering a perceptible output during a session only while an affirmation input from each participant whose affirmation the session requires is received within an affirmation window, the perceptible output being selected from playback of media content, a haptic pattern, a lighting state, an ambient tone, and actuation of a device; upon expiry of the affirmation window without such an affirmation input, changing the perceptible output gradually over a fade period; upon a stop input from any person present, changing the perceptible output abruptly in a manner perceptibly distinguishable from the gradual change; and resuming the perceptible output after either change only upon a fresh affirmation input from each participant whose affirmation the session requires.

2. A method performed by one or more devices of an enrolled participant, comprising: conducting a session in which the enrolled participant is the only participant whose affirmation the session requires and no other person present is enrolled; processing inputs attributed to the enrolled participant and discarding, without transcription, classification, or logging, speech not attributed to the enrolled participant, except that a stop input in any voice is honored; delivering prompts of the session privately to a device of the enrolled participant; upon a lapse of an affirmation window of the enrolled participant, presenting a private check-in to the enrolled participant; and, upon absence of a response to the private check-in, initiating a help response.

3. A method performed by one or more computing devices, comprising: conducting a session among one or more enrolled participants and one or more persons who are not enrolled; awaiting affirmation only from enrolled participants named in an active policy; computing a joint consent state over those enrolled participants only; continuing a perceptible output only while the joint consent state is affirmed within an affirmation window; honoring a stop input from any person present, enrolled or not; upon a lapse, changing the perceptible output and issuing a check-in; and, upon continued absence of interaction, initiating a call for help.

4. A method performed by one or more computing devices, comprising: conducting a session in which no media content is played and affirmation by each participant whose affirmation the session requires is renewed within an affirmation window; signaling the state of the affirmation window to each such participant by at least one of (a) a non-media status signal, selected from a haptic pattern on a device of the participant, a lighting state, and a low ambient tone, that continues while the window is current, changes upon a lapse, and ceases abruptly upon a stop input, and (b) no continuous signal while more than a lead time remains in the window, followed by a haptic pattern or a change of lighting of increasing insistence as the window approaches and passes expiry; and, absent an affirmation, issuing a check-in and, upon continued absence of interaction, initiating a call for help.

5. A method performed by one or more computing devices including a user device, comprising: entering a guard state upon an arming cue that is an ordinary input not naming a safety function, while carrying out the ordinary response to the arming cue; in the guard state, awaiting no affirmation and gating no output; detecting, without first detecting a wake word, a safe input of a user from a signal of at least one of a microphone, a non-acoustic speech sensor, a button, a touch sensor, and a motion sensor; upon detecting the safe input, presenting a private check-in on a device of the user; and initiating a help response upon an answer requesting help, a duress answer, or absence of an answer to the private check-in.

6. A method performed by a user device, comprising: entering a bedroom mode upon any of a bedtime signal, a determination that the user device is in a bedroom of a user, and an explicit command of the user; upon and by reason of entering the bedroom mode, detecting on the user device, without first detecting a wake word, at least one safe phrase selected from a safe phrase enrolled by the user and a universal emergency phrase adopted under a policy applicable to a locale of the user device, the set detected being enrolled safe phrases only, universal emergency phrases only, or both, as the user elects or as a local or institutional policy, a policy of the manufacturer, the copyright author, or the inventor, the terms of a software license, a community safety condition, or applicable law requires, and retaining no audio and generating no transcript of speech that does not match a safe phrase; and, upon detecting a safe phrase in the bedroom mode, executing a response comprising at least one of a check-in, an alarm, and a call for help through a pathway pre-configured by the user.

7. A method performed by a mobile computing device of a user, comprising: detecting, on the device, an arrival of the device at a place; determining, on the device, that one or more conditions are met, the conditions being selected from the place being unfamiliar by comparison with known places stored on the device, a time of the arrival falling within a configured late-

night window, a risk context being present based on at least one signal from a period preceding the arrival, the place being classified as a residence, and another device not previously associated with the user having arrived with the device; responsive to the determining, presenting a private check-in on the device or on a wearable device bound to the user, the check-in offering an all-clear answer and at least one of a greyzone answer and a help answer; and, responsive to a greyzone answer or a help answer, sending an alert to a contact the user designated in advance.

8. A method performed by a device that plays media content, comprising: playing media content through a loudspeaker during a session; capturing audio by a microphone while the media content plays; cancelling the media content from the captured audio using, as a reference, the media content that the device is causing to be played; detecting in the cancelled audio, on the device and without first detecting a wake word, a stop input and an affirmation input of each participant whose affirmation the session requires; upon detecting a stop input, stopping playback of the media content; and continuing playback only while an affirmation input of each such participant is detected within an affirmation window.

Output and media

9. The method of any one of claims 1, 3 and 8, wherein the perceptible output is playback of media content on a media playback target, and a change of the output upon a lapse comprises commanding the target to fade the media content and confirming, from a player state returned by the target, that the command took effect.

10. The method of claim 1 or claim 3, wherein the perceptible output is playback of media content and the affirmation and stop inputs are detected in microphone audio from which the media content is cancelled using a playback reference signal.

11. The method of any one of claims 1, 3 and 8, wherein the perceptible output is playback of media content and lyric or transcript data of the media content is used, with or without echo cancellation, to mask recognizer output coinciding with the lyrics, or to exclude, skip, mute, or substitute a track or segment containing an enrolled or universal word.

12. The method of any one of claims 1, 3 and 8, wherein the perceptible output is rendered audibly in a room in which the participants are present, such that its continuation, any change upon a lapse, and any change upon a stop input are each perceptible without reference to a display.

13. The method of any one of claims 1, 3 and 8, wherein the perceptible output is playback of media content, upon a lapse, a fade of the media content begins at a musical boundary at or

before the lapse, or after the lapse by no more than a maximum deferral, and, upon a stop input, the media content is stopped without regard to any musical boundary.

14. The method of any one of claims 1, 3, 4 and 8, wherein, upon a lapse, the output is changed gradually over a fade period and, upon a stop input, the output is changed abruptly or stopped within a stop latency shorter than the fade period.

Escalation and help

15. The method of any one of claims 1, 3 and 8, further comprising, after the gradual change or the lapse, upon absence of any interaction for a check-in period, issuing a check-in; upon continued absence for an alarm period, sounding an alarm; and upon continued absence for a help period, calling for help through a pre-configured pathway; wherein, before a check-in is issued, an interaction attributed to a participant halts advancement; once a check-in directed to a particular participant is issued, only a qualifying response of that participant halts advancement; after a duress input or an explicit request for help, no ordinary interaction of any person halts advancement; incidental movement is not treated as an interaction unless an active policy so provides; no interaction resumes the output; and only an affirmation resumes it.

16. The method of any one of claims 2 to 7 and 15, wherein the check-in is carried by any one or more of a haptic pattern on a wearable device, phone, or accessory, a change of a connected lighting device, a spoken prompt, and a display.

17. The method of claim 15, wherein the alarm comprises any one or more of a tone intended to be heard outside the room, flashing of a connected lighting device, continuous haptic output on a wearable device, and a spoken warning stating the time remaining before help is called.

18. The method of claim 15, wherein whether the alarm is audible in the room is elected by a participant from options including an audible alarm in the room, a silent alarm, an alarm audible only outside the room, and an alarm deferred until a responder has accepted the call for help and is within a configured distance.

19. The method of any one of claims 2 to 6 and 15, wherein the help response or call for help proceeds through one or more pathways, in an order set in advance, selected from a message or automated call to a trusted contact, a request to a peer emergency-response network, a monitoring center, a loud spoken help stage addressed to persons nearby, and a communications link to an emergency dispatch center.

20. The method of any one of claims 2 to 6 and 15, wherein the help response is tiered, first notifying a designated contact and requesting emergency services only upon at least one of an

answer requesting help, a duress answer, a corroborating distress signal, and a failure to reach any designated contact within a configured period.

21. The method of any one of claims 2 to 6 and 15, further comprising, upon acceptance of the call for help by a responder designated in advance as permitted to be admitted, and upon verification at a door that a person present is that responder, commanding an access device to grant entry for a time-limited window.

22. The method of claim 21, wherein the verification is by any one or more of a one-time code, a credential presented by a device of the responder, an access-control system of a building, a monitoring operator, and a knock pattern delivered to the responder and not rendered by any device inside the room.

23. The method of claim 21, wherein entry is granted only where a location reported by a device of the responder is within a configured distance of the door.

24. The method of claim 21, wherein unlocking is delayed for a courtesy interval announced in the room and to the responder, the interval being shortened or waived when an emergency requires.

25. The method of any one of claims 2 to 6 and 15, further comprising, upon acceptance of the call for help by a responder, rendering in the room a pre-enrolled melodic motif indicating that help is on the way.

26. The method of any one of claims 1 to 6 and 8, further comprising recognizing an enrolled melodic motif hummed, whistled, or sung by a person, by matching a key- and tempo-independent representation of its intervals, and treating it as a request for help.

27. The method of any one of claims 2 to 6 and 15, wherein an enrolled OK phrase or all-clear answer cancels pending pathways, and any pathway already invoked is notified of the cancellation.

28. The method of any one of claims 2 to 6 and 15, wherein the call for help includes a dispatchable location of the room comprising a civic address, a unit, a floor, a room identifier, and wayfinding instructions, confirmed by reading a room tag in the room.

Consent state and inputs

29. The method of any one of claims 1, 3, 4 and 8, wherein affirmation is required from two or more participants and a joint consent state is computed as the minimum of their consent states, a

stop from any one withdrawing it.

30. The method of any one of claims 1 to 4 and 8, wherein affirmation inputs are enrolled at two or more tiers including an enthusiastic tier, and an active policy specifies the tier required to renew the affirmation window.

31. The method of any one of claims 1 to 6 and 8, further comprising classifying utterances on the user device into meaning categories including at least a go category, a greyzone category, and a stop category, wherein enrolled stop and go words are honored independently of, and regardless of, the classification.

32. The method of claim 31, wherein a stop category is acted upon at a lower confidence than a go category.

33. The method of any one of claims 1 to 6 and 8, further comprising classifying a stop utterance as a bare stop, a stop with an expressed reason of pain, fear, or distress, or a stop of explicitly limited scope, and responding to each differently, an enrolled stop word pausing any gated output in every case.

34. The method of claim 33, wherein, upon a stop of explicitly limited scope in an utterance containing no enrolled stop word, any gated output is reduced without being paused and the request is announced to another participant in the speaker's words.

35. The method of any one of claims 1 to 4 and 8, further comprising, upon a greyzone input, conducting a time-limited clarifying exchange whose questions are answerable by a single input and present stopping and continuing with equal prominence.

36. The method of any one of claims 1 to 6 and 8, further comprising, upon an enrolled brake input, entering a brake state distinct from withdrawal in which any gated output is faded, the participants are prompted to separate and settle, and any gated output resumes only after each participant confirms calm and gives a fresh affirmation, the brake being recorded as a cooperative pause.

37. The method of any one of claims 1 to 6 and 8, further comprising recognizing a secret concern cue enrolled privately by a participant and bound to a mode of concern, and representing only the resulting response, not the cue or the participant, in any output perceptible to another person.

38. The method of any one of claims 1 to 6 and 8, further comprising recognizing a duress variant of an affirmation, indistinguishable to another person from an ordinary affirmation, and,

upon recognizing it, crediting no affirmation, leaving every output perceptible to another person unchanged, and silently invoking a help pathway.

39. The method of any one of claims 2 and 5 to 7, wherein the check-in accepts an enrolled duress answer whose perceptible effect is identical to that of an all-clear answer and which silently initiates the help response.

40. The method of any one of claims 1 to 4 and 8, further comprising estimating an affective quality of a participant's voice under a tone mode the participant elects for his or her own voice from at least an off mode, a private mode, and a graded mode.

41. The method of any one of claims 1 to 6 and 8, further comprising attributing inputs to a participant by an on-device voiceprint comparison, with a liveness check, for a participant who has separately consented to biometric processing, a stop input being honored at any attribution level.

42. The method of any one of claims 1 to 6 and 8, wherein an affirmation is given by a momentary biometric touch on a device enrolled by the participant, and no camera image is made available.

43. The method of any one of claims 1 to 6 and 8, wherein inputs from a wearable device, phone, or accessory bound to a participant are attributed to that participant and no other, the binding being presumed or proven, and a stop input from any device being honored regardless of binding.

44. The method of any one of claims 1 to 4 and 8, wherein an affirmation is given by a deliberate gesture sensed by a wearable device, selected from a nod sensed by an earbud and a hand clasp sensed by wearable devices of two participants whose proximity is confirmed by short-range ranging.

45. The method of any one of claims 1 to 6 and 8, wherein a wireless handheld input device, selected from a panic button, a media remote, a camera-shutter remote, a presentation clicker, and a game controller, maps press patterns to signals selected from an affirmation, a brake, a stop, a close, a concern cue, a duress affirmation, and a silent call for help.

46. The method of any one of claims 1 to 6 and 8, wherein the stop input or safe input comprises an entry of a versioned registry of universal safe words recognized without enrollment and without a wake word.

47. The method of any one of claims 1 to 6 and 8, wherein an input or output is provided through an alternative modality selected for a person with a disability and carries the same authority, timing, attribution, and logging as the modality it replaces.

48. The method of any one of claims 1 to 6 and 8, wherein at least one input is articulated silently and detected from a signal of a non-acoustic speech sensor worn by or directed at the person giving it, an affirmation so given being credited only where each participant whose affirmation the session requires has elected to allow silent affirmations.

49. The method of claim 48, wherein the non-acoustic speech sensor and the article carrying it are a pairing marked in Table 1.

50. The method of any one of claims 1 to 6 and 8, further comprising, upon a stop or emergency input of a participant, relaying speech attributed to that participant, by voiceprint or by the device or button that triggered the input, through a loudspeaker at elevated volume, speech of any other voice not being relayed.

Policy, stages, and relationship

51. The method of any one of claims 1 to 4 and 8, wherein a participant enrolls a boundary map of words, areas, and acts, held in the participant's own encrypted record, against which a scoped or greyzone input is resolved.

52. The method of any one of claims 1 to 4 and 8, further comprising maintaining a current stage of the session from an ordered plurality of stages, ascent requiring a stage-entry affirmation and descent being available without affirmation and without being treated as a withdrawal.

53. The method of claim 52, wherein a boundary between stages is set by default at the line drawn by applicable law between sexual contact and a sexual act.

54. The method of any one of claims 1 to 4 and 8, further comprising, upon wind-down cues, suspending the affirmation window and, upon departure of a participant, granting a grace period during which escalation advances no further than a private check-in, any risk indicator ending the grace period.

55. The method of any one of claims 1 to 4 and 8, wherein an active policy is derived from each participant's policy protectively, the shortest timing value governing, a stop from any participant being honored, and resumption requiring every participant's resume conditions.

56. The method of any one of claims 1, 3 and 4, wherein, in a discreet mode, the perceptible output is a haptic pattern perceptible only to the participants, so that a lapse produces no audible or visible change in the room, prompts are delivered only by haptic output, and a call for help is made silently.

57. The method of any one of claims 1 to 4 and 8, wherein the affirmation window is shortened for a pair of participants without a history of sessions together, or after an interval exceeding a target cadence, and lengthened only by agreement of both participants.

58. The method of any one of claims 1 to 4 and 8, further comprising receiving from each participant privately a role preference and suggesting role assignments only where both preferences permit, either participant declining without a reason being recorded to the other.

59. The method of any one of claims 1 to 4 and 8, further comprising maintaining privately for each participant a receptive load and weighting role suggestions toward exchange where the loads are unequal, the weighting never preventing a session or a role the participants choose.

60. The method of any one of claims 1 to 4 and 8, further comprising recording initiations, automatically or manually after the fact, and presenting an initiation balance to each participant who has elected to see it.

Records, privacy, and transport

61. The method of any one of claims 1 to 8, further comprising recording state transitions, escalation stages, and responses in a hash-chained, append-only event log without recording audio.

62. The method of claim 61, wherein each entry records an identifier and hash of each recognition model whose output produced it, and the archived model is available for a reproduction test upon a claim that an input was missed.

63. The method of claim 61, further comprising, upon an emergency condition, freezing and sealing consented session data in an encrypted evidence locker, each access being logged.

64. The method of any one of claims 1 to 8, wherein the method maintains no persistent connection to a server, opens an outbound connection only upon an emergency or a greyzone requiring review, and transmits a minimized, encrypted payload through one or more transports selected from an HTTPS request, a push notification carrying an opaque token, an end-to-end encrypted message, an oblivious relay, an SMS, MMS, or RCS message, a satellite message, and a peer-to-peer or mesh relay.

65. The method of any one of claims 1 to 4 and 8, further comprising detecting a soft close and ending the session with a single notification instead of advancing escalation.

Guard state, bedroom mode, and arrival

66. The method of claim 5, wherein the arming cue is selected from a media selection, a request to a voice assistant, a command to a connected device, a press or touch pattern on a button or wearable device, a gesture, a placement of a phone, a silently articulated phrase, and a touch message composed on a wearable device.

67. The method of claim 6, further comprising, upon detecting a device of a second person in the bedroom, offering the user, or automatically entering, a guard state.

68. The method of claim 6, wherein the determination that the user device is in a bedroom is made from any one or more of charging cycles, wake-sleep cycles, dismissal of a wake-up alarm, sleep detected by a wearable device, a place fingerprint recorded by a phone, a room tag, and an assignment of the device to a bedroom in a smart-home application.

69. The method of any one of claims 2, 5 and 6, wherein the method operates only for a user established by attestation or inference to be at or above a required minimum age, the user being treated as below it where such signals are absent or conflict.

70. The method of any one of claims 5, 6 and 48, further comprising, upon detecting that a wearable device carrying a sensor in use has been taken off, continuing with at least one remaining channel and privately notifying the user where no silent channel remains.

71. The method of claim 5 or claim 6, further comprising exposing to an artificial-intelligence assistant authorized by the user a function for invoking the private check-in and help response, the assistant invoking it in place of answering conversationally without itself requesting emergency services.

72. The method of claim 5 or claim 6, further comprising interpreting an instruction given by the user in the user's own words into actions selected from a set approved in advance, wherein no instruction cancels, delays, or downgrades a pending help response or check-in.

73. The method of claim 7, wherein the risk context is based on any one or more of a breath-alcohol reading at or above a configured level, an on-device inference of a drinking episode, a departure from a drinking occasion, an indication of a date, co-travel of another device, a ride that ended at the place, a night-out mode, a request from a chosen contact, and an impairment flag.

74. The method of claim 7, wherein a place is unfamiliar when it matches no known place whose familiarity statistics, comprising distinct days of presence or cumulative dwell time over a trailing period, exceed a threshold, familiarity optionally being judged at the level of a floor estimated from barometric pressure.

75. The method of claim 7, wherein a place whose classification as a residence is uncertain is treated as a residence.

76. The method of claim 7, further comprising, responsive to the greyzone answer, offering on the device any one or more of arming a guard state, a pretext call or message, a ride, and a request for a refuge for the night.

77. The method of claim 7, wherein absence of an answer does not by default raise an alarm, and the user selects in advance among taking no action, repeating the check-in, notifying the designated contact, and advancing through an alarm and a help stage.

Solo-mode silent affirmation

78. The method of claim 2, wherein the enrolled participant renews the affirmation window by an affirmation articulated silently and detected from a signal of a non-acoustic speech sensor worn by or directed at the enrolled participant, the silent affirmation being credited upon the enrolled participant's own election without any election by another person present; wherein playback of media content continues only while the affirmation window is current, its volume following a current consent score of the enrolled participant; and wherein a fade of the media content upon a lapse or a falling consent score prompts the enrolled participant to give at least one of a spoken input, a pause, a brake, and a concern cue of a yellow or red mode, and, absent a response, advances an escalation sequence that sends a distress alert.

Voice relay

79. A method performed by one or more computing devices, comprising: upon a trigger by a first person, the trigger being selected from a stop input, a brake input, a concern cue, a button pattern, an enrolled megaphone word, a request in the first person's own words, and a silently articulated input, opening a voice relay; while the voice relay is open, recognizing on a device an instruction spoken or silently articulated by the first person and attributed to the first person, and relaying no speech of any other person; and rendering through a loudspeaker or to a communications endpoint a restatement of the instruction that differs from the first person's utterance in at least one of amplitude, voice, and phrasing.

80. The method of claim 79, wherein the restatement is rendered at an elevated volume, and the volume is increased while the voice relay remains open, upon repetition of the instruction, or upon sensing that the instruction has not been followed.

81. The method of claim 79, wherein the restatement is rendered in a synthesized voice having characteristics selected in advance by the first person from at least pitch, timbre, apparent gender, apparent age, accent, cadence, and firmness, or in a voice the first person chose or recorded in advance.

82. The method of claim 79, wherein the form of the restatement is selected in advance from verbatim repetition, a paraphrase in plain, direct language, a summary condensed to the operative request, and a restatement as a clear, firm instruction naming the action requested and phrased to make compliance more likely, the restatement adding no request the first person did not make.

83. The method of claim 79, wherein the instruction is articulated silently by the first person and detected from a signal of a non-acoustic speech sensor that is worn by or directed at the first person and senses movement or activity of articulators of the first person, and the restatement voices the instruction aloud.

84. The method of claim 79, wherein the first person is identified by at least one of voiceprint attribution, an enrolled device, wearable, or button that raised the trigger, and a non-acoustic speech sensor worn by or directed at the first person.

85. The method of claim 79, further comprising, where a recognition confidence is below a threshold, asking the first person a confirming question privately through an earpiece, wearable, or phone, relaying only a confirmed instruction, and advancing an escalation sequence to a help stage upon a confirmed request for help.

86. The method of claim 79, wherein the voice relay operates during at least one of a consent session in which an output continues only while affirmation is renewed, a guard state, and a bedroom mode, and each restatement is preceded by a distinctive tone.

87. The method of claim 83, wherein at least one of: the instruction is articulated with the lips closed; the trigger is an articulatory cue enrolled by the first person and selected from a pattern of presses of the tongue, a silent tongue click, a pattern of taps of the teeth, and a jaw clench held for a set time or repeated in a pattern; and the non-acoustic speech sensor and the article carrying it are a pairing marked in Table 1.

Voice relay, continued

88. The method of claim 79, wherein recognizing the instruction attributed to the first person comprises isolating the first person's subsequent speech by at least one of receiving the instruction from a near-mouth or contact sensor of the first person, verifying each utterance against an enrolled speaker model selected upon the trigger, tracking the first person's position with a microphone array, and accepting an utterance only within an instruction window opened by the first person and indicated privately to the first person, the utterance also being verified against an enrolled speaker model or received from a personal sensor of the first person; and wherein an ambiguous or overlapping segment is not relayed.

89. The method of claim 79, further comprising supplying the restatements rendered by the voice relay, and any media being played, as references for echo cancellation, so that the voice relay does not relay its own output or the playback of another device.

90. The method of claim 79, wherein the restatement is produced by decoding the instruction into fields comprising an action, an addressee, a scope, and a polarity; generating a candidate restatement; verifying that the candidate preserves the action, the scope, and the polarity and contains no demand or factual assertion absent from the instruction; and, where the verification fails, rendering an approved literal form, asking the first person privately, or rendering the instruction verbatim.

91. A method performed by one or more computing devices, comprising: upon a trigger by a person, the trigger being selected from a button press pattern and a universal safe phrase, rendering through a loudspeaker or to a communications endpoint a fixed announcement that a stop or help has been requested, the announcement restating none of the person's words and requiring no speaker model or personal sensor of the person to be enrolled.

92. The method of claim 80, wherein the volume is increased in steps up to a configured maximum only upon repetition of the instruction by the first person or upon evidence that the instruction has not been followed, the evidence being selected from continued proximity sensed by ranging, contact corroborated by a bed sensor or a wearable device, and a further stop, brake, or concern input, movement alone not being treated as such evidence.

93. The method of claim 79, wherein the voice relay operates independently of any consent session and renders the restatement through at least one of a phone speaker, a remote smart speaker, and a call to a communications endpoint.

94. The method of claim 79, wherein the trigger further includes a duress input, but only where the first person specifically elected in advance that a duress input open the voice relay.

Components, states, handoff, and delivery

95. A wearable device comprising: a non-acoustic speech sensor positioned by the device to sense movement or activity of articulators of a wearer; a processor; and a wireless interface; the processor being configured to detect, from a signal of the sensor and without first detecting a wake word, a safety input articulated silently by the wearer and selected from a stop input, a brake input, a concern cue, a duress input, a request for help, and an affirmation; to generate a safety event comprising an event type, a pseudonym of the wearer, a counter, a time stamp, and a confidence, signed by a key held on the device; and to transmit the safety event to a controller without transmitting the signal of the sensor.

96. A method performed by an output controller, comprising: receiving safety events from one or more input devices, each event comprising an event type, a participant pseudonym, a counter, a time stamp, and a signature; verifying the signature and rejecting an event whose counter or time stamp is not fresh; and, responsive to a verified event, changing an output under the control of the output controller according to the event type, the output controller implementing no sensor from which the events were derived.

97. A method performed by one or more computing devices, comprising: maintaining an internal safety state and, separately, a presented state that governs every output perceptible to a person other than a user; upon a duress input of the user, recording duress in the internal safety state, crediting no affirmation, and invoking a help pathway, while holding the presented state as it would be for an ordinary input unless the user elected in advance that a duress input produce an overt announcement or voice relay; upon an enrolled stop input, changing both the internal safety state and the presented state, a duress input given together with a stop input being recorded internally as both while the presented state shows only the stop; and recording each divergence between the states in a sealed log.

98. The method of claim 97, wherein the user has made the election, and the presented state produces the overt announcement or voice relay upon the duress input.

99. A method performed by one or more computing devices, comprising: monitoring a user for safety inputs through a first input channel; upon determining that the first input channel will cease, requesting a second input channel to report readiness, operating the first and second input channels concurrently for an overlap period while correlating events of the two channels by participant, event type, and overlapping time window under a canonical event identifier, and releasing the first input channel only after the second input channel has reported readiness; upon determining that the first input channel has ceased without such a handoff, entering a degraded-

protection state, activating a replacement input channel where one is available, and recording the interval without coverage; maintaining a pending check-in, an affirmation timer, and an escalation stage independently of any input channel; and, where no input channel remains, privately notifying the user.

100. A method performed by one or more computing devices, comprising: generating an emergency event having an event identifier; transmitting the event through a first transport and, absent confirmation of delivery within a timeout, through a second transport with the same event identifier; tracking the event through states of queued, transmitted, delivered, accepted, and resolved; rendering an indication that help is on the way only upon the accepted state; and, upon a cancellation by the user after the accepted state, transmitting a cancellation referencing the event identifier without withdrawing the accepted request from the responder.

Arming and response alternatives

101. A method performed by one or more computing devices including a user device, comprising: entering a guard state upon at least one of an explicit command of a user, a covert arming cue that is an ordinary input not naming a safety function, and automatically upon a context selected from a bedroom mode, an arrival check-in, and detection of a device of a second person; in the guard state, awaiting no affirmation and gating no output; detecting, without first detecting a wake word, a safe input of the user; and, upon detecting the safe input, executing a response configured for the safe input.

102. A method performed by one or more computing devices including a user device, comprising: detecting, without first detecting a wake word, a safe input of a user that belongs to a help category; and, responsive thereto, at least one of initiating a help response without an intervening check-in and taking a local safety action without contacting any person, the local safety action being selected from stopping media, a pretext call or message, and a voice-relay announcement.

Recognition of silent inputs

103. The method of any one of claims 48, 78 and 83, wherein the silently articulated input is recognized by at least one of comparing a segmented signal with templates of cues enrolled by the user, subject to a per-cue rejection threshold set from the user's ordinary movements recorded during calibration; classifying the signal into a closed vocabulary of safety intents, an input outside the vocabulary producing no action other than a private check-in; and decoding the signal into text or phonetic units with a confidence, a decoding below a confidence threshold never cancelling or downgrading a pending check-in or help response.

104. The method of claim 103, wherein each of a plurality of sensor families has a sensor-specific front end producing an embedding at a regular rate, and the embeddings feed a shared intent model.

105. The method of any one of claims 48, 78 and 83, wherein a detection is attributed to the person wearing or targeted by the non-acoustic speech sensor only where it is consistent with that person's enrolled patterns, external motion sensed by an inertial sensor is treated as an indicator of interference, a detection coinciding with such motion being acted upon only upon corroboration or after a private check-in, and an ambiguous detection results in a private check-in.

Sensing and interpretation policy

106. A method performed by one or more computing devices, comprising: presenting to a participant a selector by which the participant selects a position or any combination of individually selectable privacy features, each position or combination being mapped to a processing policy that specifies which sensors and recognizers operate on inputs of that participant, which categories of words, phrases, and articulatory cues are recognized, and whether transcription, semantic interpretation, and affect analysis are permitted; upon a selection, determining the permissions that the selection removes, adds, and leaves unchanged; revoking removed permissions at once; previewing added permissions without activating them and activating them only upon authorization by each participant whose inputs would newly be processed; propagating a versioned policy to each participating device and awaiting, within a bounded timeout, an acknowledgment from each that it applied the policy; and presenting each device's configuration as confirmed or unconfirmed, separately from the requested configuration.

107. The method of claim 106, wherein the selectable positions or features include separate choices for silently articulated inputs, selected from no articulation recognition, enrolled articulatory cues only, enrolled silent phrases, a limited safety vocabulary, and general silent-speech decoding, and the selector indicates which input channels remain active for each selection.

108. The method of claim 106, wherein, where participants make different selections, the processing of each participant's inputs is limited to that participant's selection by at least one of separate personal input channels, attribution of each segment to a participant before any more expansive processing stage, and, where reliable separation is unavailable, limiting a shared microphone to the most restrictive selection made by any participant whose speech it may capture.

109. The method of claim 106, wherein recording, transmission, biometric attribution, and affect analysis are each authorized separately from the selected position, and authority to cancel a pending help response is separate from the processing policy.

110. The method of claim 106, wherein each selection corresponds to a preset bundle of functions, to an independent choice of sensing, interpretation, retention, and sharing, or to a combination of features selected by checkboxes, toggles, or a grid of features by participant or device, and each preset is associated with an allowed-egress manifest against which actual transmissions are compared.

111. The method of claim 106, wherein a device that cannot comply with the policy, or that does not acknowledge it within the timeout, is placed in a configuration that remains within the processing authorizations then in force, even where a safety function consequently becomes unavailable, the reduced protection being shown; wherein a device that cannot be reached enforces locally held authorizations that expire unless renewed; and wherein authorization of a session-scoped selection reverts to a resting selection when the session ends.

Transcription only upon a conflict signal

112. The method of any one of claims 1 to 8, 79 and 106, wherein, until a conflict signal is detected, inputs are recognized only by on-device keyword spotting, template matching, or classification into a closed set of events, none of which generates a transcript, and no audio or sensor signal is retained beyond a rolling buffer held in memory; and wherein, upon a conflict signal selected from a stop input, a brake input, a concern cue, a distress indicator, a duress input, a request for help, a greyzone answer, and a trigger of a voice relay, at least one of a conflict recording, diarization, and transcription begins, as permitted by an active policy and applicable law.

113. The method of claim 112, wherein the diarization or transcription is limited to participants whose selected policy permits it, ends upon resolution of the conflict or expiry of a configured period, whichever is first, and is held encrypted, each participant being able to decrypt only the content that participant is authorized to receive.

Systems and media

114. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of any one of claims 1 to 94 and 96 to 113.

115. The system of claim 114, further comprising a non-acoustic speech sensor configured to be worn by or directed at a user.

116. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform the method of any one of claims 1 to 94 and 96 to 113.

ABSTRACT OF THE DISCLOSURE

Rules of interpretation, separability statements, and pairing tables are provided for a consent and safety system in which a perceptible output, such as music, continues only while each required participant renews an affirmation, fades upon a lapse, and stops abruptly upon a stop input, with escalation through a check-in and an alarm to a call for help. Each feature may be used alone or in combination. This includes standalone solo, unenrolled-partner, no-media, and guard-state modes; a bedroom mode; an arrival check-in triggered by any one or more contextual conditions; safety inputs given aloud, by touch, by button, or by silent articulation sensed by a non-acoustic speech sensor; a voice relay that restates a distressed person's spoken or silently articulated instructions louder, in a chosen voice, or more clearly; and a selector whose chosen sensing and interpretation policy is enforced and confirmed on each device.